

# Contemporary Models of Development and Underdevelopment

## Chapter 4

Econ 322 – The Global Economy: Trade and Development

University of Tennessee, Knoxville

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Based on Todaro, P. M. and Smith, C. S. (2020), *Economic Development*, 13th ed., Pearson.

# Overview

## Transition to Contemporary Development Theory

### Key Insight:

- ▶ After **50+ years** of development experience, we now understand:
  - ▶ Development is **possible**, but also **extremely difficult**
  - ▶ It does **not happen automatically**
  - ▶ Requires **systematic effort, policy, and institutional capacity**

### Why New Models Emerged:

- ▶ **Classic theories** provided valuable ideas (e.g., capital, structural change, dependency)
- ▶ But new challenges and experiences (e.g., **East Asia success, sub-Saharan Africa struggles**) demanded better tools
- ▶ Result: A wave of **formal, refined models** since the **late 1980s**

# Outline

- ▶ **Underdevelopment as a Coordination Failure**
- ▶ Multiple Equilibria: A Diagrammatic Approach
- ▶ Starting Economic Development: The Big Push
- ▶ Michael Kremer's O-Ring Theory of Economic Development
- ▶ Economic Development as Self-Discovery
- ▶ The Hausmann-Rodrik-Velasco Growth Diagnostics Framework

# Underdevelopment as a Coordination Failure (1/9)

## ► Historical Origins

- **Paul Rosenstein-Rodan (1943):** *Big Push Theory* – large-scale coordinated investment is essential for development.
- **1950s–1970s Structuralist Economists:** emphasized indivisibilities, inter-sectoral linkages, and public investment. Example figures: Ragnar Nurkse, Albert Hirschman.
- **1980s–2000s Formalization:** use of **game theory** and **multiple equilibria** to explain stagnation. Key models:
  - Cooper & John (1988): macro coordination games
  - Murphy, Shleifer & Vishny (1989): formal big push
  - Kremer (1993): O-Ring model
- **Broader Influence:** linked to endogenous growth theory & institutional economics; highlights complementarity, trust, and credible commitment.

## Underdevelopment as a Coordination Failure (2/9)

**Complementarity:** an action taken by one firm, worker, or organization that increases the incentives for other agents to take similar actions. Complementarities often involve investments whose return depends on other investments being made by other agents.

**Coordination failure:** a state of affairs in which the inability of agents to coordinate their behavior (choices) leads to an outcome (equilibrium) that leaves all agents worse off than in an alternative situation that is also an equilibrium.

No firm invests in a remote area because no one else has, and no one else invests for the same reason: a low-level equilibrium trap that is hard to escape without external coordination.

# Underdevelopment as a Coordination Failure (3/9)

- ▶ **Coordination failures**
- ▶ Coordination failures can occur **even when agents have full information**.
  1. People have **diverse expectations** about others' behavior.
  2. It's often rational to **wait for someone else to act first**.
- ▶ **Hypothetical Example: College friends in Ankara, Dec 31st, 2020**
  - ▶ All want to meet, all know others are in town, but no coordination on the location.
  - ▶ Result: Some go to Tunalı, some to Armada, others to Eymir → **no one meets**.

## Underdevelopment as a Coordination Failure (4/9)

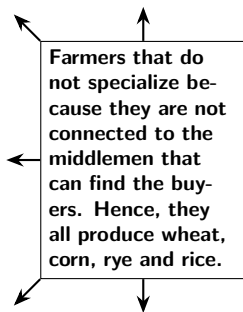
### ► Coordination failures: Firms–Workers Example

| Firms                      | Workers              |                             |
|----------------------------|----------------------|-----------------------------|
|                            | Invest in new skills | Do not invest in new skills |
| Invest in new tech.        | (100, 100)           | (−100, 0)                   |
| Do not invest in new tech. | (0, −100)            | (0, 0)                      |

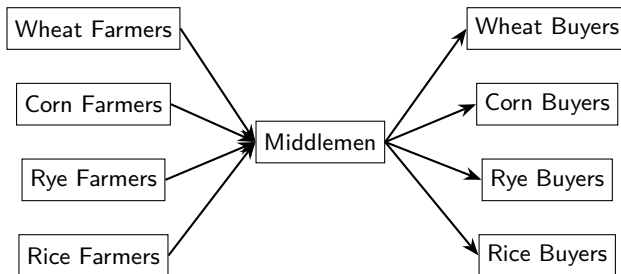
Each cell shows (Firms' payoff, Workers' payoff).

# Underdevelopment as a Coordination Failure (5/9)

## ► Coordination failures: Farmers–Middlemen–Buyers Example



### Specialized Farmers





# Underdevelopment as a Coordination Failure (6/9)

## Real-World Examples

- ▶ **Manufacturing in Sub-Saharan Africa:** No firms enter because there is no infrastructure; the government doesn't invest in infrastructure because there are no firms.
- ▶ **Education and labor markets in Latin America:** Parents hesitate to invest in schooling when there are no skilled jobs; firms avoid creating skilled jobs because workers are untrained.
- ▶ **Women's employment in South Asia:** Social norms and employer biases mutually reinforce low female labor participation.



Jurong Industrial Estate, Singapore: a state-planned industrial park where roads, power, and plants were built together.

*Wikimedia Commons, Calvin Teo, CC BY-SA 2.5*

# Underdevelopment as a Coordination Failure (7/9)

- ▶ **Governments' role in removing the coordination failures**
- ▶ **Deep Intervention:** A government policy that can shift the economy to a **better equilibrium** or higher **long-run growth path**, which becomes **self-sustaining** over time.  
→ Once this better equilibrium is reached, continued policy enforcement may no longer be needed.
- ▶ **Example:** Simultaneous **subsidies** to both firms and workers:
  - ▶ Firms: Incentivize investment in **new technologies**
  - ▶ Workers: Incentivize investment in **new skills**
- ▶ *Note:* Easier said than done — success depends on political economy, governance quality, and institutional capacity.

## Underdevelopment as a Coordination Failure (8/9)

- ▶ **Cooperation VS. Competition**
- ▶ In some settings, it is more beneficial for firms or individuals to **cooperate** rather than **compete**.
- ▶ Think of a **repeated prisoner's dilemma**: each round, two parties compete, but both would earn higher payoffs if they consistently cooperate.
- ▶ However, sustaining cooperation is difficult:
  - ▶ It requires **patience** (long-term perspective)
  - ▶ It also requires **mutual trust**
- ▶ **Key Question:** How can we design institutions or incentives that promote cooperation over time?

# Underdevelopment as a Coordination Failure (9/9)

- ▶ **Critiques and Evaluation**

- ▶ **Strengths**

- ▶ Provides an explanation for persistent poverty despite the availability of technology
- ▶ Supports the case for targeted government interventions to shift the economy toward better equilibria

- ▶ **Limitations**

- ▶ Empirical identification of multiple equilibria is difficult
- ▶ There is a risk of government failure or corruption
- ▶ Coordination does not guarantee efficiency — it may only lead to a relatively better, but still suboptimal, outcome

# Outline

- ▶ Underdevelopment as a Coordination Failure
- ▶ **Multiple Equilibria: A Diagrammatic Approach**
- ▶ Starting Economic Development: The Big Push
- ▶ Michael Kremer's O-Ring Theory of Economic Development
- ▶ Economic Development as Self-Discovery
- ▶ The Hausmann-Rodrik-Velasco Growth Diagnostics Framework

# Multiple Equilibria: A Diagrammatic Approach (1/12)

- ▶ **Historical Origins**
- ▶ The concept of **multiple equilibria** dates back to **Keynesian economics**, where underemployment can be a stable outcome.
- ▶ **Development economists** like **Rosenstein-Rodan** and **Murphy, Shleifer & Vishny (1989)** formalized it in “Big Push” models for industrialization.
- ▶ In **game theory**, multiple equilibria arise in **coordination games**, where agents’ decisions depend on each other.
- ▶ **Cass and Shell (1983)** introduced **sunspot equilibria** in macroeconomics, where outcomes are driven by self-fulfilling beliefs.

## Multiple Equilibria: A Diagrammatic Approach (2/12)

- ▶ **Core Idea**
- ▶ Agents' optimal actions **depend on expectations** of others' behavior.
- ▶ When decisions are **strategic complements**, then:

*“The more I believe others will act, the more I want to act too.”*

- ▶ This interdependence can lead to **multiple equilibria**, depending on how the **best-response function** is shaped.

# Multiple Equilibria: A Diagrammatic Approach (3/12)

## Multiple equilibria

A condition in which more than one equilibrium exists. These equilibria may sometimes be ranked, in the sense that one is preferred to another, but the unaided market will not move the economy to the preferred outcome.

### Example: Equilibria with higher growth versus higher equality

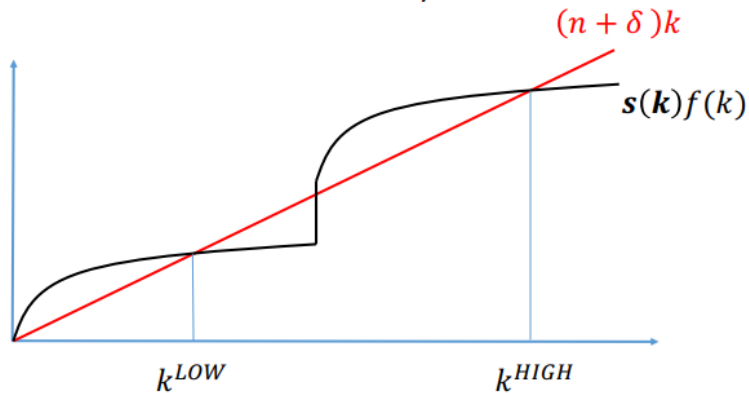
- ▶ An equilibrium with high growth but large inequality, or one with more equality but slower growth
- ▶ Neither outcome is guaranteed: the one that prevails depends on beliefs, institutions, and whether mechanisms exist to shift expectations

Why might markets fail to move from a bad equilibrium to a better one, even if everyone prefers it? Each individual waits for others to move first; no one wants to take the risk alone.

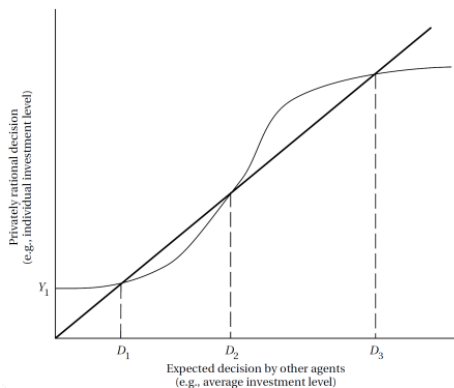


## Multiple Equilibria: A Diagrammatic Approach (4/12)

**Example:** A Solow Model with Two Steady-States



# Multiple Equilibria: A Diagrammatic Approach (5/12)



Vertical axis,  $Y_i$  Private investment of firm  $i$

Horizontal axis,  $X$  Average investment

$Y_i$  changes with  $X$  but in a nonlinear way:  $Y_i = F(X)$

- ▶ When average is low, firm  $i$ 's investment is lower.
- ▶ When average is getting larger, firm  $i$  increases its investment level.
- ▶ There is a critical level of  $X$ , denoted by  $D_2$ , at which firm  $i$ 's growth of investment is largest.

By definition, we have  $X = \mathbf{E}(Y)$ , where  $\mathbf{E}$  is the expectation operator.

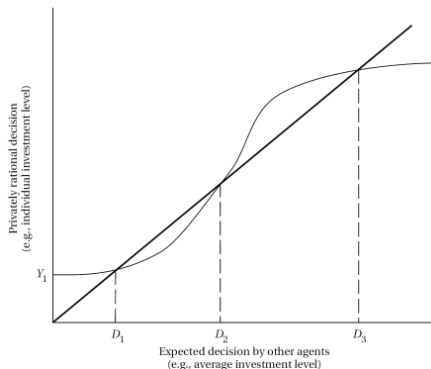
Symmetric Equilibrium:  $Y^* = F(\mathbf{E}(Y^*))$

# Multiple Equilibria: A Diagrammatic Approach (6/12)

There are three equilibria that satisfy  $Y^* = F(E(Y^*))$ :  
 $Y^* = D_1$ ,  $Y^* = D_2$ ,  $Y^* = D_3$ .

However, only  $D_1$  and  $D_3$  are stable. In the former, there is low investment, and it is consistent with the expectation of low investment. In the latter, there is high investment. Again, expectations are consistent with this outcome.

Why is  $D_2$  unstable? If we deviate to a point slightly larger than  $D_2$  (i.e.,  $E(Y) > D_2$ ), private investment at that point is larger than  $D_2$ . Hence, private investment grows until it reaches  $Y^* = D_3$ . Similarly, if we deviate to a point slightly smaller than  $D_2$  (i.e.,  $E(Y) < D_2$ ), private investment at that point is smaller than  $D_2$ . Hence, it starts decreasing until it reaches  $Y^* = D_1$ .



## Multiple Equilibria: A Diagrammatic Approach (7/12)

Let's construct another example of multiple equilibria:

$Y_i$  # of videos YouTuber  $i$  uploads to the public domain in one week

$X$  average # of videos YouTubers upload to the public domain in one week

Clearly, if  $X$  is low, it is not much enjoyable to spend time on YouTube (not much content to see or not many people to interact with). However, at increasing levels of  $X$ , more YouTubers can join and add more videos. Then there are two stable equilibria as in the figure.

This applies to many other sectors of the economy where private decisions are affected by the average behavior in the economy/society.

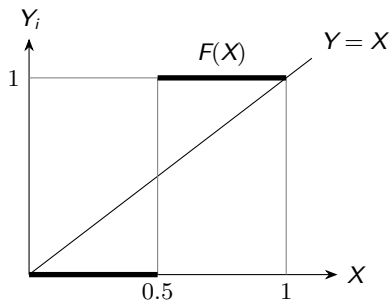
## Multiple Equilibria: A Diagrammatic Approach (8/12)

We can imagine an example of interpersonal trust:

$Y_i \in [0, 1]$  Citizen  $i$ 's interpersonal trust to others (higher  $Y_i$ , higher trust level)

$X \in [0, 1]$  average level of interpersonal trust in the society

$$Y_i = F(X) = \begin{cases} 0, & X < 0.5 \\ 1, & X \geq 0.5 \end{cases}$$



**Find the equilibria.**

**Discuss the stability of each equilibrium point.**

# Multiple Equilibria: A Diagrammatic Approach (9/12)

## Ranking multiple equilibria

In models of multiple equilibria, it is typical to think of welfare rankings.

Is one of the equilibria better than the others in the sense of **Pareto**?

Put differently, if a government intervenes to the economy and moves the equilibrium from one to the other, does this intervention increase at least one agent's welfare without decreasing the welfare of any other agent?

# Multiple Equilibria: A Diagrammatic Approach (10/12)

## History versus Expectations

- ▶ In one class of models with multiple equilibria, **how the economy chooses one equilibrium over the others is determined solely by initial conditions, i.e., by history**. In such models, there is typically no information friction or any role defined for expectations (The economy cannot jump from one equilibrium to the other).
- ▶ In another class, models feature an information structure, and expectations may matter. For instance, even though the initial conditions are not favorable for the Pareto-dominating equilibria, **expectations may dominate history** if they could be coordinated for the most-preferred equilibrium (The economy jumps to this equilibrium).

# Multiple Equilibria: A Diagrammatic Approach (11/12)

## History versus Expectations

- ▶ **Paul Krugman** has a very nice, intuitive paper about such models.
- ▶ You may also search the JSTOR database for papers that use such reasoning in understanding economic dynamics.
- ▶ Some relevant concepts are:
  - ▶ Self-fulfilling prophecy
  - ▶ Jump variables
  - ▶ Perfect foresight equilibrium
  - ▶ “History matters!”
  - ▶ Indeterminacy



# Multiple Equilibria: A Diagrammatic Approach (12/12)

- ▶ **Evaluation**
- ▶ **Contributions:**
  - ▶ Explains why similar countries follow **divergent development paths**
  - ▶ Offers a theoretical basis for **escaping low-level traps**
  - ▶ Encourages **nonlinear, threshold-based policy thinking**
- ▶ **Critiques:**
  - ▶ **Empirical identification** of multiple equilibria is difficult
  - ▶ **Equilibrium outcomes** are sensitive to model assumptions
  - ▶ Relies heavily on **government coordination ability**, which may be weak or absent in practice

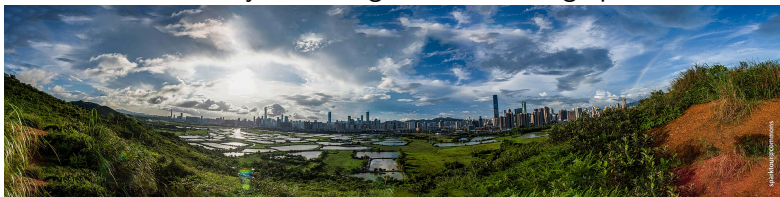
# Outline

- ▶ Underdevelopment as a Coordination Failure
- ▶ Multiple Equilibria: A Diagrammatic Approach
- ▶ **Starting Economic Development: The Big Push**
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# Starting Economic Development: The Big Push (1/15)

## The Idea of the Big Push

- ▶ How can an economy that was stagnant for long centuries initiate **modern growth**?
- ▶ Does the **prolonged stagnation** affect the likelihood of a successful growth takeoff?
- ▶ Does modern growth **sustain** itself in the long run once it is somehow initiated?
- ▶ Interesting example is **Argentina**; its earlier growth record signalled good prospects, but the economy later stagnated for a long episode.



Shenzhen's skyline seen from Hong Kong: a small border town in 1980, a megacity today — a take-off that was coordinated rather than automatic.

# Starting Economic Development: The Big Push (2/15)

## The Idea of the Big Push

- ▶ According to **Paul Rosenstein-Rodan**'s model, initiating industrialization is difficult in an agricultural (subsistence) economy because no one wants to be the first industrialist!
- ▶ Why?
- ▶ Imagine a subsistence economy with a possibility of introducing a new industrial good.
- ▶ Suppose that there is no exporting. Then, who will buy the goods produced by the first firm to industrialize?
- ▶ Since workers keep consuming existing subsistence goods, the profitability of the firms in the newly industrialized sector depends on how many firms there are to produce the new good.

# Starting Economic Development: The Big Push (3/15)

## The Idea of the Big Push

- ▶ The **profitability** of one factory depends on whether another one opens, which in turn depends on its own potential profitability, and that in turn depends on the profitability of still other factories.
- ▶ Such **circular causation** should now be a familiar pattern of a coordination failure problem.
- ▶ Besides, there is also the problem of training workers for the new technology. A firm that trains workers will create a positive **externality** for the other firms.
- ▶ But since training workers is costly, no one wants to be the **first entrepreneur** that bears the cost.

# Starting Economic Development: The Big Push (4/15)

## The Graphical Representation

There are six assumptions:

1. **Factors:** Labor is the only factor of production. The total supply is  $L > 0$  and is fixed.
2. **Sectors:** There are two sectors: modern and traditional. In the traditional sector, the wage is normalized to 1. In the modern sector, the wage is denoted by  $W$  and is larger than 1 ( $W > 1$ ).
3. **Technologies:** There exists a large number of products,  $N$ . In the traditional sector, there is constant returns to scale. In the modern sector, there is increasing returns to scale. Any good can be produced with traditional and modern technologies.

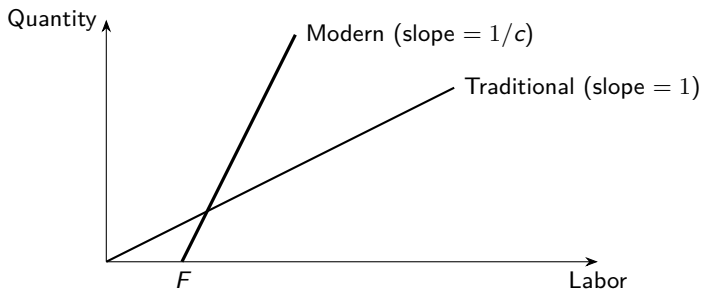
# Starting Economic Development: The Big Push (5/15)

## The Graphical Representation

More specifically, the modern sector production function is

$$Q = \frac{L - F}{c}, \quad c < 1, F > 0$$

where  $F > 0$  is the minimum number of workers for the production to start (the fixed cost) and  $c < 1$  is the marginal cost.



# Starting Economic Development: The Big Push (6/15)

## The Graphical Representation

The other three assumptions:

4. **Demand:** An equal share of income is spent on different products. Hence, if  $Y$  is national income, the money spent on each good is  $Y/N$ . There is no saving.
5. **International trade:** For simplicity, it is assumed in this version that the economy is closed.
6. **Market structure:** Traditional sector is perfectly competitive. The linear production function  $Q = L$  and  $W = 1$  implies  $P = 1$ .

$$TR = PQ - WL = PL - L = (P - 1)L$$



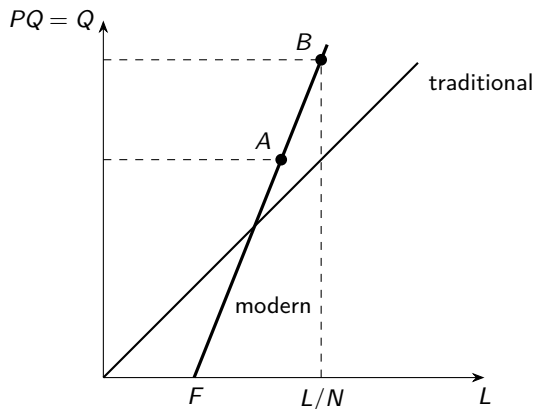
# Starting Economic Development: The Big Push (7/15)

## The Graphical Representation

6. **Market structure (continued):** In the modern sector, for each good, there is only one monopoly firm that cannot charge a price larger than one because of the competitive pressure of traditional producers.

**Remark:** Under these assumptions, the profitability of a modern technology firm in any one good depends on whether the other goods in the economy are produced with traditional or modern technologies. If more of the goods are produced by modern technology, then this creates larger demand for other goods through higher wages.

## Starting Economic Development: The Big Push (8/15)



**Note:** *This graph is for one of the sectors.*

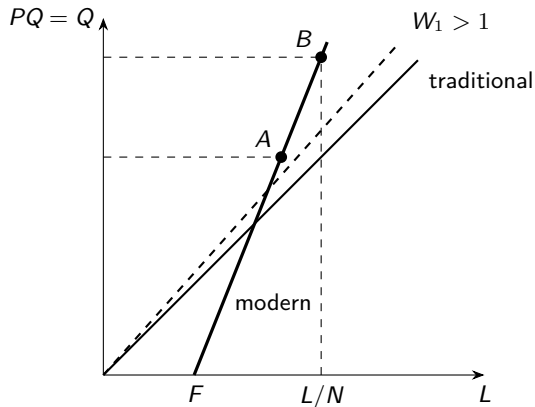
► **Point A**

If a modern firm enters but there are traditional firms in the other sectors of the economy, the modern firm operates at point A.

► **Point B**

If a modern firm enters but there is no traditional firm in the economy, then the modern firm operates at point B.

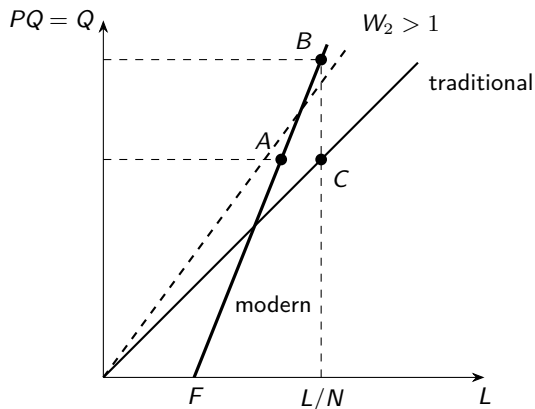
## Starting Economic Development: The Big Push (9/15)



- ▶ **Case 1:**  $W_1 > 1$
- ▶ The modern firm would find it profitable to operate at **A** since its profit is strictly greater than zero when the modern sector wage is equal to  $W_1$ .
- ▶ Since this applies to the market for any other good, each and every good is produced by modern firms.
- ▶ Then equilibrium is at point **B** since no traditional firms remain. All firms industrialize.

**No Need for a Big Push**

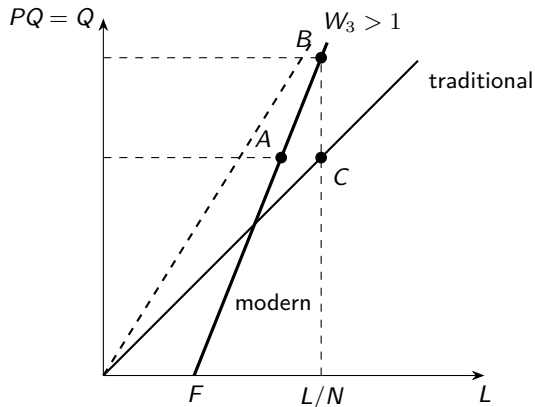
## Starting Economic Development: The Big Push (10/15)



- ▶ **Case 2:**  $W_2 > 1$
- ▶ The modern firm is not profitable to operate at **A** under this higher wage level.
- ▶ Modern firms would enter if all goods are produced by modern firms (at point **B**)
- ▶ Two equilibria are possible:
  1. All goods are produced with traditional technology (**C**)
  2. All goods are produced with modern technology (**B**)

**There is a need for a Big Push that coordinates modern firms to enter the market for all goods.**

## Starting Economic Development: The Big Push (11/15)



- ▶ **Case 3:**  $W_3 > 1$
- ▶ Since the modern sector wage is now very high, no modern firm would enter even if they all enter simultaneously.
- ▶ Hence, even if there is a Big Push that can coordinate the modern firms to enter all markets, cost conditions are not favorable for a profitable operation of the modern technology.
- ▶ **The economy cannot industrialize, and the unique equilibrium is at point C.**

# Starting Economic Development: The Big Push (12/15)

## The Graphical Representation

### Summary:

1. If modern sector wages are low enough, then there is no need for a Big Push since all goods are produced by modern firms (unique eq.)
2. If modern sector wages are higher but not too high, then there is a need for a Big Push. The profitability of a modern firm depends on whether other goods are produced by other modern firms. There are then two equilibria: (i) all goods are produced with traditional tech. versus (ii) all goods are produced with modern tech.
3. If modern sector wages are sufficiently high, then modern sector technology is not profitable even if all goods are produced with modern tech. Then, there is no industrialization (unique eq.)

# Starting Economic Development: The Big Push (13/15)

## Further Issues

In the baseline model studied in this lecture, the firms are connected mainly through demand. In an extended model, we could also think of technological externalities.

- ▶ **Technological externality:** A positive or negative spillover effect on a firm's production function through some means other than market exchange.
- ▶ *Example:* Firms learning new production methods from one another in a costless way.

# Starting Economic Development: The Big Push (14/15)

## Further Issues

### Other Cases in Which a Big Push May Be Necessary

1. **Intertemporal effects:** Even if the modern sector wage is also equal to one, there can still be multiple equilibria if there are intertemporal effects. For instance, if the investment that makes production more efficient in period  $(t + 1)$  must be undertaken in period  $(t)$ , then firms must expect that demand will be large enough in period  $(t + 1)$ , which in turn requires modern firms invest simultaneously.
2. **Urbanization effects:** In a model with rural versus urban areas, the urban demand may be concentrated more on the modern sector (i.e., manufacturing). Hence, it may be necessary to push urbanization to achieve industrialization.



# Starting Economic Development: The Big Push (15/15)

## Further Issues

### Other Cases in Which a Big Push May Be Necessary

3. **Infrastructure effects:** Infrastructure investments by some firms create a positive externality for other firms that benefit from such infrastructure investments to decrease their fixed costs. But this does not imply that infrastructure is sufficient to push for industrialization.
4. **Training effects:** There is underinvestment in training by entrepreneurs because they expect that the workers they trained may find employment in rival firms. Besides, there is little demand for training by the workers since they do not know what to learn.

# Outline

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- ▶ Starting Economic Development: The Big Push
- ▶ **Michael Kremer's O-Ring Theory of Economic Development**
- ▶ Economic Development as Self-Discovery
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# Michael Kremer's O-Ring Theory of Economic Development (1/11)

## The Model

- ▶ Modern production (especially in contrast to traditional crafts production) requires that **many activities be done well together** in order for any of them to amount to high value.
- ▶ **Challenger** space shuttle exploded in 1986 as a result of failure of one small inexpensive part called the O-ring.
- ▶ The model builds upon the notion of **strong complementarity** of production inputs (in technical terms, such production functions are called *supermodular*).



Michael Kremer (b. 1964), author of the O-ring theory (1993); Nobel Prize 2019.

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# Michael Kremer's O-Ring Theory of Economic Development (2/11)

## The Model

- ▶ The model is at the firm level (but it can easily be extended to the macro level).
- ▶ Consider a firm among many others.
- ▶ Let  $y$  denote the output of the firm.
- ▶ Suppose that, for the firm to produce the good or service, there are  $n$  different tasks to be completed.
- ▶ Output depends on the “quality”  $q \in (0, 1)$  of each task.
- ▶ Formally, we have

$$y = q_1 \times q_2 \times q_3 \times \cdots \times q_n$$

# Michael Kremer's O-Ring Theory of Economic Development (3/11)

## The Model

- ▶  $q_i$  has several different interpretations:
  - ▶ It can denote the skill level of worker who is assigned to the task
  - ▶ It can denote the probability of success for the task
  - ▶ It can denote any other measurable dimension of input qualifications

**Example:** Imagine a single firm whose production structure requires two tasks ( $n = 2$ ). The firm has four workers with two different skill levels. Specifically, two workers have the **high skill** level  $H$ , and two workers have the **low skill** level  $L < H$ .

# Michael Kremer's O-Ring Theory of Economic Development (4/11)

## The Model

**Example (continued):** In one production cycle, two workers are assigned to each of the tasks. Hence, we have

$$y = q_i q_j$$

- ▶ Here,  $q_i$  denotes the skill level of the worker who is assigned to the task  $i$ , and  $q_j$  denotes the skill level of the worker who is assigned to the task  $j$ .
- ▶ The question is of course how the firm should allocate its four workers to two different tasks in two production cycles.

# Michael Kremer's O-Ring Theory of Economic Development (5/11)

## The Model

### Example (continued):

- ▶ **Allocation 1:** High-skill workers are matched with low-skill workers. Then, the total production is

$$y = HL + HL = 2HL$$

- ▶ **Allocation 2:** High-skill workers are matched with high-skill workers, and low-skill workers are matched with low-skill workers. Then, the total production is

$$y = HH + LL = H^2 + L^2$$

- ▶ Clearly, we have  $H^2 + L^2 > 2HL$  since  $(H - L)^2 > 0$ .

# Michael Kremer's O-Ring Theory of Economic Development (6/11)

## The Model

### Example (continued):

- ▶ Allocation 2 allows the firm to produce more. This is the result of strong complementarity. This type of matching (HH and LL) is called **positive assortative matching**. This means that workers with high skills will work together and workers with low skills will work together.

How can we understand the role of the O-ring model as a model of development and poverty??



# Michael Kremer's O-Ring Theory of Economic Development (7/11)

## The Model

### How can we understand the role of the O-ring model as a model of development and poverty??

- ▶ If high-skill workers are matched with high-skill ones or if high-tech firms hire high-skill workers, then such assortative matching processes increases inequality and duality within an economy.
- ▶ Why? Because high-skill agents benefit from strong complementarities in terms of wages, and low-skill agents are harmed.
- ▶ Besides, these effects may contribute to regional inequality within a country. High-tech firms located together in a region with high-skill people.

# Michael Kremer's O-Ring Theory of Economic Development (8/11)

## The Model

### Another Example:

- ▶ Suppose that there are 2 firms and 2 workers in the economy. The production functions are

$$y(1) = A(1) h(1) \qquad y(2) = A(2) h(2)$$

where  $y$  is output,  $A$  is a firm's productivity, and  $h$  is a worker's skill level when employed.

- ▶ Suppose that the workers' skill levels are equal to  $2x$  and  $6x$  (one of the workers is 3 times more skilled).
- ▶ Suppose that the firms' productivity levels are equal to  $A$  and  $7A$  (one of the firms is 7 times more productive).

# Michael Kremer's O-Ring Theory of Economic Development (9/11)

## The Model

### Another Example (continued):

- ▶ Positive assortative matching requires high productivity firm to be matched with high-skill worker (and vice versa).
- ▶ Then, the output levels of firms in the economy is

$$y(i) = A \times 2x \qquad y(j) = 7A \times 6x$$

- ▶ Then, the total output in the economy is  $44(Ax)$ . If the assortative matching was not pursued, the total output would be  $20(Ax)$ .

# Michael Kremer's O-Ring Theory of Economic Development (10/11)

## The Model

### Yet Another Example:

- ▶ Suppose that there two strongly complementary tasks.
- ▶ Task 1 is always performed by a human worker, but Task 2 can be performed with either a human worker or a robot worker.

$$y = h \times (h + \beta r)$$

- ▶ Here,  $h$  denotes the number of hours supplied by the human workers, and  $r$  denotes the number of hours supplied by the robot worker.
- ▶  $\beta > 1$  is the productivity of the robot worker.

# Michael Kremer's O-Ring Theory of Economic Development (11/11)

## The Model

### Yet Another Example (continued):

- ▶ The firm's profit is

$$\pi = p[h(h + \beta r)] - w(2h) - p_r r$$

- ▶ Here,  $p$  is the product's price,  $w$  denotes the wage paid to human workers, and  $p_r$  denotes price of the robot.
- ▶ **Question.** Is there a solution to the firm's profit maximization problem?

# Outline

- ▶ Underdevelopment as a Coordination Failure
- ▶ Multiple Equilibria: A Diagrammatic Approach
- ▶ Starting Economic Development: The Big Push
- ▶ Michael Kremer's O-Ring Theory of Economic Development
- ▶ **Economic Development as Self-Discovery**
- ▶ The Hausmann-Rodrik-Velasco Growth Diagnostics Framework

# Economic Development as “Self-Discovery” (1/6)

- ▶ **What Is the “Self-Discovery” Theory?**
- ▶ Ricardo Hausmann and Dani Rodrik proposed the “Self-Discovery” theory.
- ▶ It argues that developing countries do **not know in advance** which goods or industries they are truly good at producing (i.e., where their comparative advantage lies).
- ▶ Discovering this requires entrepreneurial experimentation: investing, trying, and sometimes failing.
- ▶ However, the discovery process faces key problems:
  - ▶ It is **risky** and costly;
  - ▶ The costs are **sunk costs** (i.e., unrecoverable);
  - ▶ Once a firm discovers a profitable activity, **others can copy it**, so the original firm may not fully benefit.
- ▶ Result: Markets will underinvest in self-discovery, and structural transformation is delayed.

## Economic Development as “Self-Discovery” (2/6)

- ▶ **Historical Background**
- ▶ In the mid-to-late 20th century, many developing countries pursued **industrialization**:
  - ▶ Imported technology from advanced economies;
  - ▶ State-led initiatives: building steel plants, car factories, and even airplane projects.
- ▶ But many projects **failed**:
  - ▶ The issue wasn't a lack of effort or investment;
  - ▶ The real problem was **misguided industrial choices**.
- ▶ These countries didn't know what their **real comparative advantage** was. They were shooting in the dark.
- ▶ This created large debts, inefficient industries, and disillusionment with industrial policy.



# Economic Development as “Self-Discovery” (3/6): Bangladesh’s Garment Sector

- ▶ In the early 1980s, **Bangladesh** had almost **no garment exports**
- ▶ One firm, with support from Korea’s Daewoo, experimented with **shirt exports**; the firm succeeded
- ▶ Other firms **imitated** the model
- ▶ Result: an entire **export garment industry** was born
- ▶ Today, the **ready-made garment (RMG)** sector is the backbone of Bangladesh’s economy
- ▶ Nobody knew beforehand that Bangladesh had a comparative advantage in shirts — it was **discovered**



Sewing floor of a ready-made garment factory in Dhaka, Bangladesh.

*Wikimedia Commons, Tareq Salahuddin, CC BY 2.0*

## Economic Development as “Self-Discovery” (4/6)

- ▶ In simple models, we assume that agents have
  - ▶ Perfect information
  - ▶ Symmetric information
- ▶ In reality, firms, individuals, governments (and countries as a whole) face **information** problems concerning their comparative advantages.
  - ▶ For a firm, which products to specialize in?
  - ▶ For an individual, which profession to pursue?
  - ▶ For a government, what particular policy to implement?

Development is not just a matter of deploying known inputs but a process of discovery: figuring out what to produce and how to compete in global markets.

## Economic Development as “Self-Discovery” (5/6)

- ▶ In the presence of such information problems, agents must **discover (or learn)** their comparative advantages.
- ▶ But this is difficult; there is a possibility of a **market failure**.
- ▶ Imagine some firms. They need to self-discover which particular industrial product is most advantageous to specialize in and export.
- ▶ For such a learning process, they must spend valuable resources such as time, expert labor, etc. Suppose that one firm would bear these costs and invest in self-discovery.
- ▶ The problem is that other firms, later on, benefit from the self-discovery made by this early-mover firm. Because of this **information externality**, the tragedy of commons type of reasoning implies that no firm would invest in self-discovery.

## Economic Development as “Self-Discovery” (6/6)

The three **building blocks** of the self-discovery argument:

1. Uncertainty about what products a country can produce efficiently
2. Costly local adaptation of imported technology (labor force, materials, institutions)
3. Rapid imitation after these obstacles have been overcome, reducing the incentive to be the pioneer

Case examples include India, Bangladesh, Pakistan, South Korea and Colombia: South Korea did not start off knowing it would dominate in semiconductors, shipbuilding, or electronics.

Development is not just about investing in more capital or improving education — it is about figuring out what your economy is good at. That process often requires active policy support to overcome the initial market failure.

# Outline

- ▶ Underdevelopment as a Coordination Failure
- ▶ Multiple Equilibria: A Diagrammatic Approach
- ▶ Starting Economic Development: The Big Push
- ▶ Michael Kremer's O-Ring Theory of Economic Development
- ▶ Economic Development as Self-Discovery
- ▶ **The Hausmann-Rodrik-Velasco Growth Diagnostics Framework**

# The Hausmann-Rodrik-Velasco Growth Diagnostics Framework

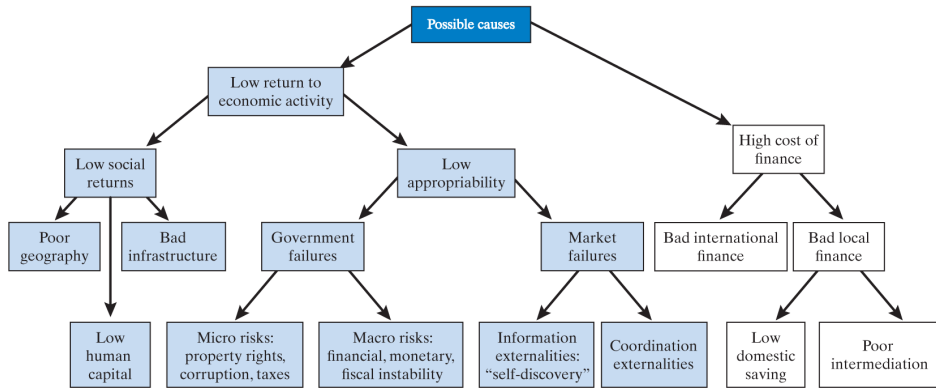
- ▶ **Focus on a country's most binding constraints on economic growth** — not all constraints, just the one or two actually holding back investment today
- ▶ **No “one size fits all” in development policy** — policies that worked in South Korea or Chile might fail in Ghana or Nepal because the binding constraints differ
- ▶ **Requires careful research to determine the most likely binding constraint** — a diagnostic tool, not a formula

Instead of asking “what is missing in this country?”, growth diagnostics asks: “what is the **single most binding constraint** on economic growth right now?” — the constraint that, if relaxed, would produce the biggest jump in private investment.

Like a good doctor: ask questions, run tests, and identify the root cause of the illness before prescribing medicine.

# Hausmann-Rodrik-Velasco Growth Diagnostics Decision Tree

*Problem: Low levels of private investment and entrepreneurship*



Start from “low private investment” and work down to the leaf boxes.

*Source: Hausmann, Ricardo, Rodrik, Dani and Velasco, Andrés (2006), ‘Getting the diagnosis right,’ Finance and Development 43. Reprinted with permission.*

# Growth Diagnostics: Categories of “Most Binding Constraints”

- ▶ Constraint categories are the boxes from which no further arrows emanate – **10 are included**
- ▶ These 10 are fairly comprehensive; many specific constraints fit well within these categories
  - ▶ Example: **natural disasters** pose immediate constraints, but if extended problems for the growth follow, **government failure** is likely the root problem, not the initial shock of the disaster
- ▶ But potential additional independent “boxes” for constraint types that have been raised for consideration but are debatable could include:
  - ▶ **Environment**
  - ▶ Inability to resolve conflict
  - ▶ Social capital (or “cultural”) dimensions such as lack of trust (e.g. in forming business relationships, etc)



# Doing Growth Diagnostics in Practice\*

- ▶ In practice, growth diagnostics involves “economic detective work.”
- ▶ To evaluate whether a proposed constraint is binding, a growth diagnostician looks for evidence on its implications.
- ▶ If the constraint is excessive taxation, we would expect high movement into the informal sector or underground economy
- ▶ If the constraint is infrastructure, we would expect significant congestion
- ▶ If the constraint is education, expect high rates of return to education
- ▶ In general, the analyst will *look for economic behavior consistent with agents trying to get around a constraint*
- ▶ *\*Draws from Ricardo Hausmann, Bailey Klinger, Rodrigo Wagner, Doing Growth Diagnostics in Practice: A ‘Mindbook,’ CID Working Paper No. 177, 2008*

## Hausmann et al's\* "Principles of a Differential Diagnosis"

- ▶ If a constraint is binding, then...
- ▶ The (shadow) price of the constraint should be high
- ▶ Movements in the constraint should produce significant movements in the objective function
- ▶ Agents in the economy should be attempting to overcome or bypass the constraint
- ▶ Agents less intensive in that constraint should be more likely to survive and thrive, and vice versa
- ▶ *\*Ricardo Hausmann, Bailey Klinger, Rodrigo Wagner, Doing Growth Diagnostics in Practice: A 'Mindbook,' CID Working Paper No. 177, 2008*

# Inclusive Growth Diagnostics: The Case of Bangladesh (1/5)

## Introducing Bangladesh

- ▶ Densely populated South Asian country bordered by India and the Bay of Bengal
- ▶ GDP growth averaged over 6% a year for much of the past two decades; poverty rates cut in half since the 1990s
- ▶ Known globally for its textile and garment industry, a major source of exports and employment, especially for women
- ▶ Remaining challenges: energy shortages, limited access to finance, urban congestion, social inequality



Bangladesh and its neighbours.

## Inclusive Growth Diagnostics: The Case of Bangladesh (2/5)

- ▶ Growth diagnostics methods were used to identify the binding constraint(s) for growth in Bangladesh
- ▶ A study goal: beyond growth as a general objective; consider *inclusive growth* in the context of opportunities for women.
- ▶ General review ruled out some potential constraints, narrowed focus to five plausible ones. Candidate binding constraints:
- ▶ Finance: Not a strong case, but recommended the finance sector would need attention looking ahead
- ▶ Market failure (particularly coordination and information externalities): Found market failures important concerns, but unlikely currently binding given severity of other problems

# Inclusive Growth Diagnostics: The Case of Bangladesh (3/5)

## ▶ Human capital:

- ▶ Progress, including girls' education, noted;
- ▶ Returns to education are not particularly high;
- ▶ Firms did not cite education levels as a problem;
- ▶ Education does not explain international migration patterns;
- ▶ No clear relationship between labor intensity and growth by industry
- ▶ While education is not now binding, the study argued that its low average quality and high variance could make it a possible emerging constraint going forward

# Inclusive Growth Diagnostics: The Case of Bangladesh (4/5)

- ▶ Government failure?
  - ▶ Evidence for contract enforcement as binding was inconclusive
  - ▶ Corruption: Authors termed it “a most binding constraint”; “close to binding” in some sectors; likely not binding on growth at this time, based on benchmark indexes and firm surveys
  - ▶ Note: Amount of bribes paid is substantial, but around the middle of comparator countries- far below Cambodia and Philippines; no clear relationship between corruption and investment.
- ▶ Infrastructure?

## Inclusive Growth Diagnostics: The Case of Bangladesh (5/5)

- ▶ The authors termed electricity infrastructure “The most binding constraint.”
  - ▶ Regarding a high shadow price, the team reported that the cost of non-grid electricity is several times that of bulk electricity tariff rates
  - ▶ Regarding movements in the constraint shifts objective, they found that electricity availability has been followed (likely causally) by higher electricity-using investment
  - ▶ Regarding bypassing constraints, firms and the government – using expensive private generators, also connected to the building of export processing zones
  - ▶ (Regarding whether less-intensive users of electricity thrive, evidence less clear)
  - ▶ Argued that greater electricity capacity and reliability would likely lead to increased formal sector employment
- ▶ Author's conclusion: electricity capacity investment would have the largest impact on growth, with favorable impacts on employment and incomes for women

## Box 4.2: Village Coordination and Monitoring for Better Health (1/2)

- ▶ The problem: poor health outcomes, low public health care provision in rural Uganda
- ▶ Villagers have access to public “dispensaries,” but use levels and satisfaction are low; government oversight has been ineffective
- ▶ A program intervention provided villagers with the knowledge and resources to enable them to monitor health workers, both individually and through their community organization
- ▶ 50 villages selected; 25 for the “treatment,” and 25 for the control group
- ▶ Surveys were conducted of all households on health outcomes and satisfaction with the health care providers; Data were also collected from health care providers
- ▶ A randomized controlled trial (RCT) was used to identify the causal impacts of the intervention



## Box 4.2: Village Coordination and Monitoring for Better Health (2/2)

- ▶ Purpose of an RCT: It is impossible to observe the same person with and without treatment...
- ▶ Next best: Find the average effect on a group of people randomly selected for treatment, from a broader, randomly selected pool. Those not selected become the control group
- ▶ Treatment: Using household survey data, a report card for each health care provider was developed and presented at community meetings, with health care providers present
- ▶ Goal: giving information to users about overall and relative outcomes enables them to more effectively monitor their health care providers
- ▶ Effects: significant improvements in health outcomes
- ▶ Source: M. Bjorkman and J. Svensson, "*Village Coordination and Monitoring for Better Health Outcomes*, Quarterly Journal of Economics, 124 (2), pp 735–769, May 2009

## Chapter summary

- ▶ **Coordination failure:** when actions are complementary, rational agents can get stuck in an equilibrium that leaves everyone worse off than another feasible equilibrium (firms vs. workers, farmers vs. middlemen); a **deep intervention** can shift the economy to the better, self-sustaining equilibrium.
- ▶ **Multiple equilibria:** when the best response rises with the average action and crosses the 45-degree line more than once, the economy has stable low and high equilibria; history and expectations decide which one prevails.
- ▶ **Big push** (Rosenstein-Rodan, Murphy-Shleifer-Vishny): a single modern firm may be unprofitable while modern firms in all sectors together are profitable; wages, intertemporal, urbanization, infrastructure, and training effects can all make a coordinated push necessary.
- ▶ **O-ring theory** (Kremer): with multiplicative production  $y = q_1 q_2 \cdots q_n$ , workers of like skill are matched together (positive assortative matching), high-skill workers and firms cluster in the same regions, and one weak link can ruin high-value output.
- ▶ **Self-discovery** and **growth diagnostics** (Hausmann, Rodrik, Velasco): countries must discover their comparative advantage, and policy should relax the single most binding constraint — electricity in the Bangladesh case.

# Concepts for Review

- ▶ Agency costs
- ▶ Asymmetric information
- ▶ Big push
- ▶ Complementarity
- ▶ Congestion
- ▶ Coordination failure
- ▶ Deep intervention
- ▶ Economic agent
- ▶ Growth diagnostics
- ▶ Information externality
- ▶ Linkage
- ▶ Middle-income trap
- ▶ Multiple equilibria
- ▶ O-ring model
- ▶ O-ring production function
- ▶ Pareto improvement
- ▶ Pecuniary externalities
- ▶ Poverty trap
- ▶ Prisoners' dilemma
- ▶ Social returns
- ▶ Technological externality
- ▶ Underdevelopment trap
- ▶ Where-to-meet dilemma